CSA1604-DATA WAREHOUSING AND DATA MINING LAB EXPERIMENTS

Day 1

List of Programs

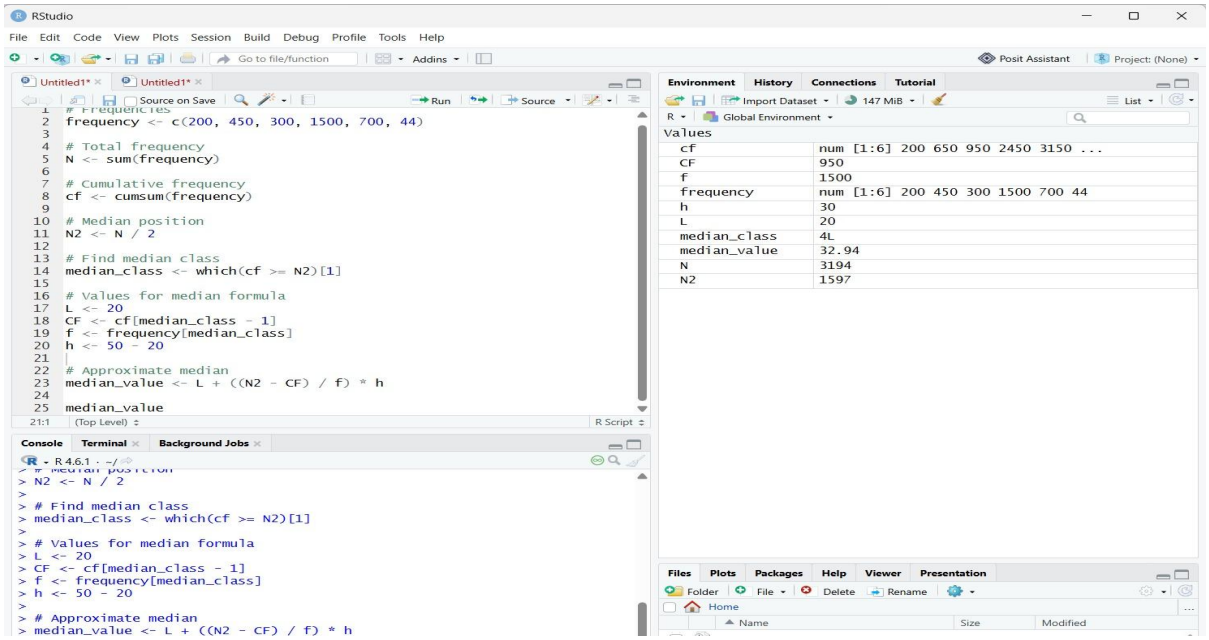
1.The intervals and corresponding frequencies are as follows. age frequency
1-5. 200
5-15 450
15-20 300
20-50 1500
50-80 700
80-110 44
Compute an approximate median value for the data

AIM:
To calculate the approximate median of grouped age-frequency data.

ALGORITHM:

1. Calculate the total frequency.
2. Find N/2N/2.
3. Identify the median class using cumulative frequency.
4. Apply the grouped median formula.
5. Obtain the approximate median.

OUTPUT:



RESULT:
The approximate median is **32.94 years**.

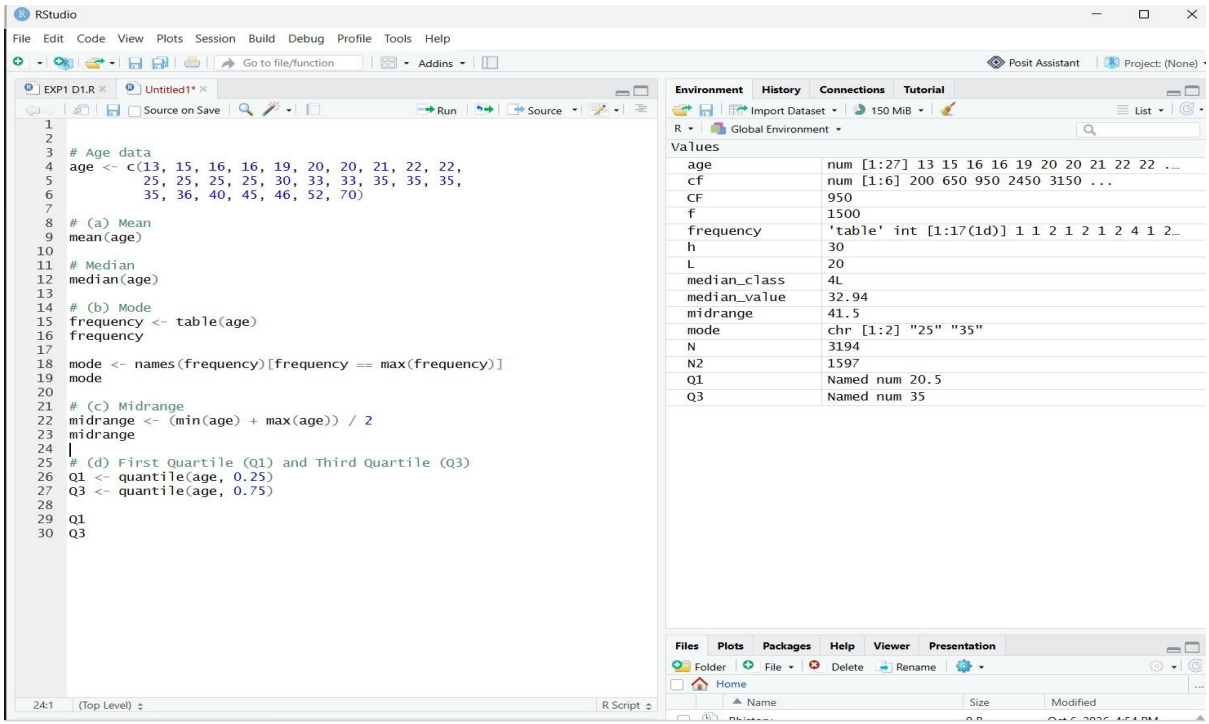
2.Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.
(a) What is the mean of the data? What is the median?
(b) What is the mode of the data? Comment on the data's modality (i.e., bimodal, trimodal, etc.).
(c) What is the midrange of the data?
(d) Can you find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data?

AIM:
To calculate the mean, median, mode, midrange, Q1 and Q3 of the given age data.

ALGORITHM:

1. Enter the age values in R.
2. Use mean() to find the mean.
3. Use median() to find the median.
4. Use frequency counting to find the mode.
5. Calculate the midrange using minimum and maximum values.
6. Use quantile() to find Q1 and Q3.
7. Identify the modality from the frequency values.

OUTPUT:



RESULT:

- Mean = **29.96**
- Median = **25**
- Mode = **25 and 35**
- Data is **bimodal**
- Midrange = **41.5**
- Q1 ≈ **20**
- Q3 ≈ **35**

3.Data Preprocessing :Reduction and Transformation

Use the two methods below to normalize the following group of data: 200, 300, 400, 600, 1000 (a) min-max normalization by setting min = 0 and max = 1 (b) z-score normalization

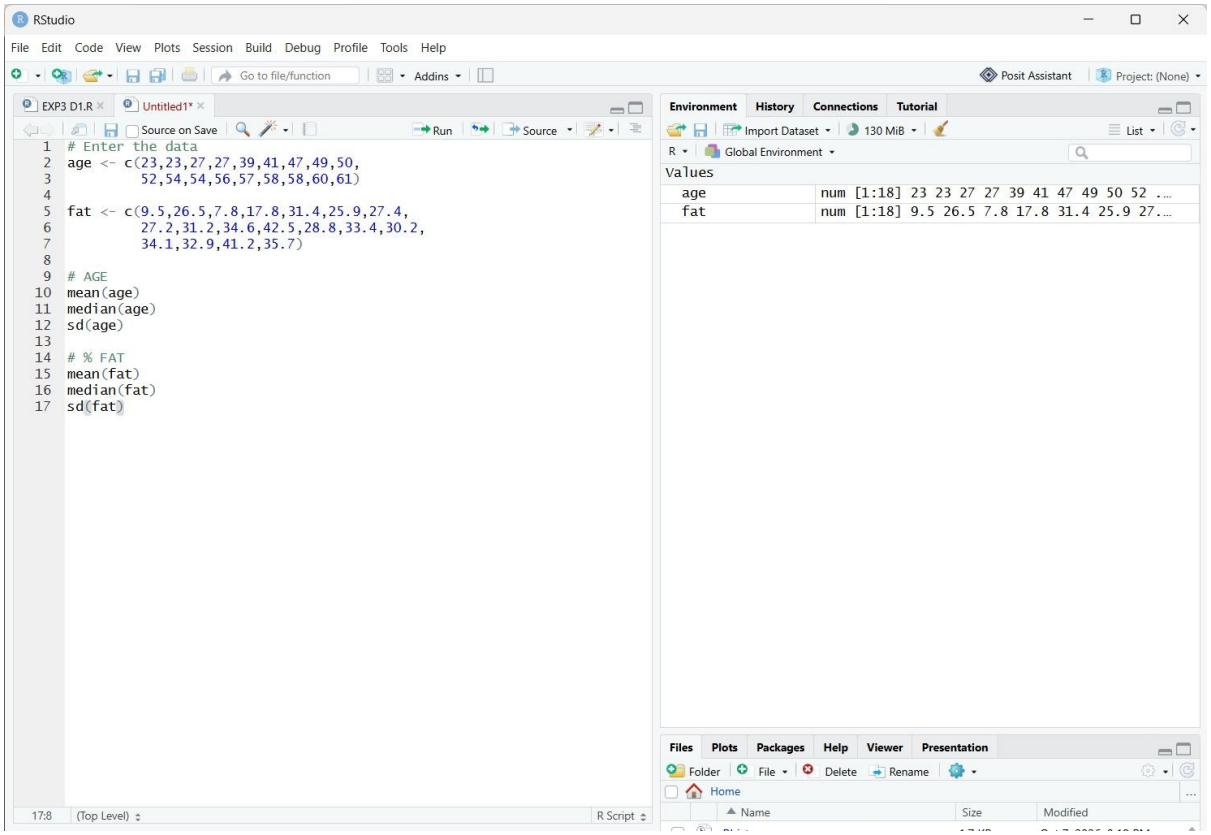
AIM:

To perform min-max and z-score normalization on the given data.

ALGORITHM:

1. Enter the data in R.
2. Find the minimum and maximum values.
3. Apply min-max normalization.
4. Calculate the mean and standard deviation.
5. Apply z-score normalization.
6. Display the normalized values.

OUTPUT:



RESULT:
The given data was successfully normalized using **min-max normalization and z-score normalization**.

4.Data:11,13,13,15,15,16,19,20,20,20,21,21,22,23,24,30,40,45,45,45,71,
72,73,75

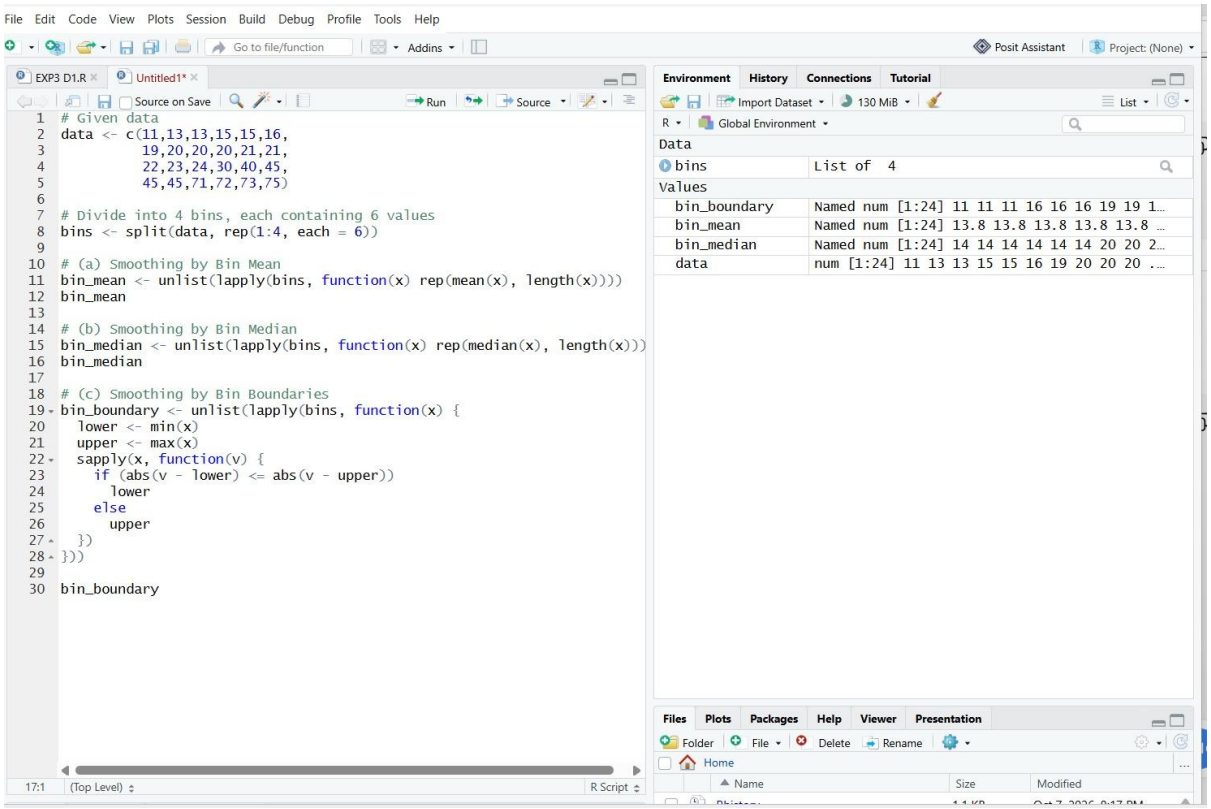
a) Smoothing by bin mean
b) Smoothing by bin median
c) Smoothing by bin boundaries

AIM:
To perform data smoothing using bin mean, bin median and bin boundaries.

ALGORITHM:

1. Enter the sorted data in R.
2. Divide the data into equal-sized bins.
3. Calculate the mean of each bin and replace values with the mean.
4. Calculate the median of each bin and replace values with the median.
5. Find the boundaries of each bin.
6. Replace values with the nearest bin boundary.
7. Display the smoothed data.

OUTPUT:



RESULT:
The data was successfully smoothed using **bin mean, bin median and bin boundary methods**.

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results:

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

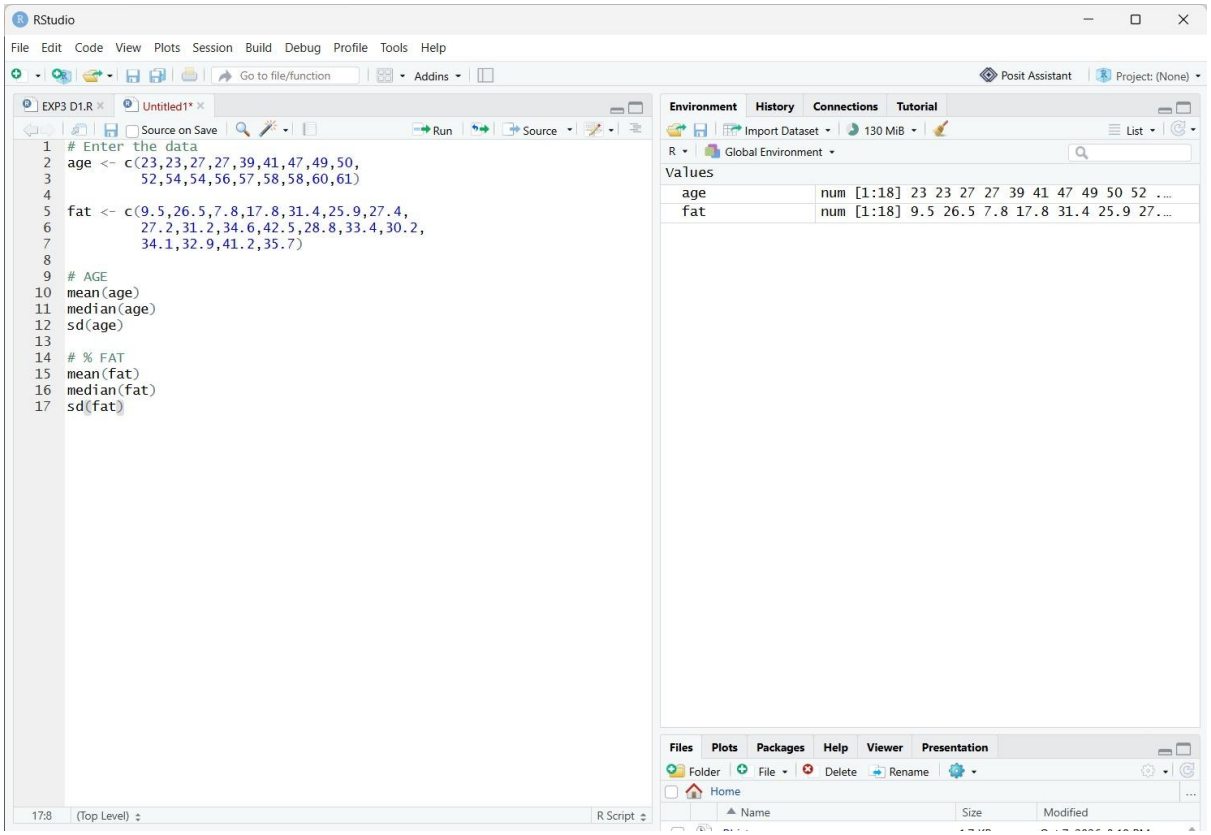
(a) Calculate the mean, median, and standard deviation of age and %fat. (b) Draw the boxplots for age and %fat.
(c) Draw a scatter plot and a q-q plot based on these two variables.

AIM:
To calculate the mean, median and standard deviation and draw boxplots, scatter plot and Q-Q plot for age and body fat.

ALGORITHM:

1. Enter age and body-fat values in R.
2. Calculate mean, median and standard deviation.
3. Draw boxplots for age and body fat.
4. Draw a scatter plot between age and body fat.
5. Draw a Q-Q plot.
6. Observe the distributions and relationship between the variables.

OUTPUT:



RESULT:
The mean, median and standard deviation were calculated successfully, and the **boxplots, scatter plot and Q-Q plot** were generated successfully.

6.Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results:

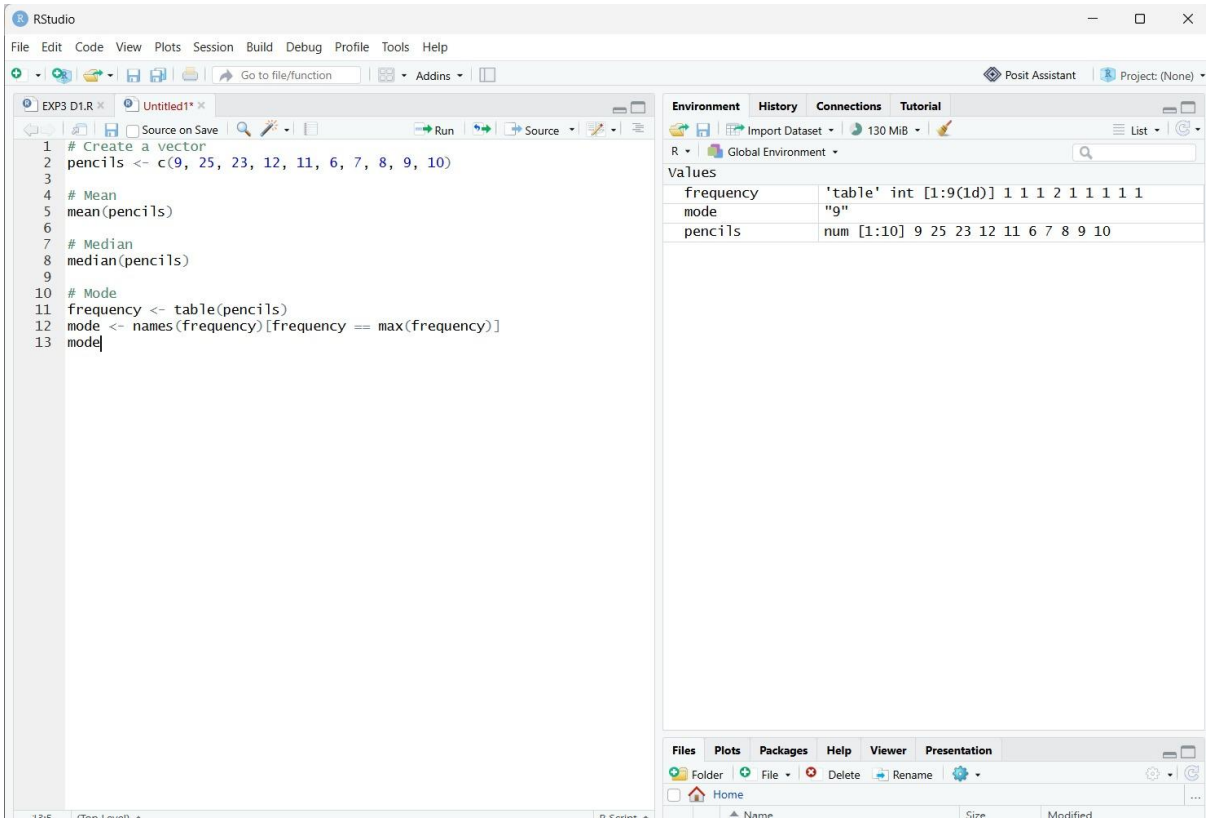
(i) Use min-max normalization to transform the value 35 for age onto the range [0.0, 1.0].
(ii) Use z-score normalization to transform the value 35 for age, where the standard deviation of age is 12.94 years.
(iii) Use normalization by decimal scaling to transform the value 35 for age. Perform the above functions using R – tool

AIM:
To perform min-max, z-score and decimal-scaling normalization for the age value 35 using R.

ALGORITHM:

1. Enter the age data in R.
2. Apply min-max normalization.
3. Apply z-score normalization using the given standard deviation.
4. Apply decimal-scaling normalization.
5. Display the normalized values.

OUTPUT:



RESULT:
The age value **35** was successfully normalized using all three normalization methods.

- Min-Max = **0.3158**
- Z-Score ≈ **-0.884**
- Decimal Scaling = **0.35**

7.The following values are the number of pencils available in the different boxes. Create a vector and find out the mean, median and mode values of set of pencils in the given data.

Box1 Box2 Box3 Box4 Box5 Box6 Box7 Box8 Box9 Box 10

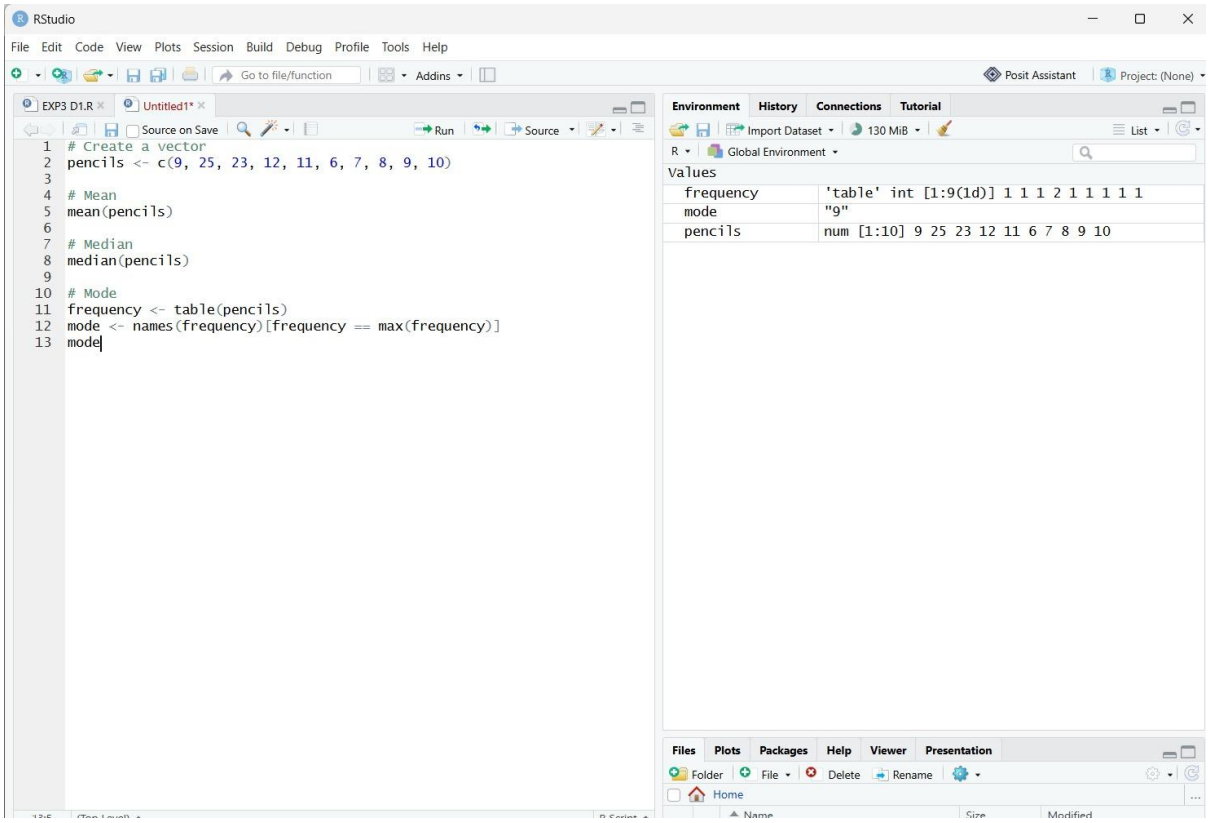
9 25 23 12 11 6 7 8 9 10

AIM:
To calculate the mean, median and mode of the number of pencils using R.

ALGORITHM:

1. Create a vector containing the pencil values.
2. Use mean() to calculate the mean.
3. Use median() to calculate the median.
4. Use frequency counting to determine the mode.
5. Display the results.

OUTPUT:



RESULT:

- Mean = **12**
- Median = **9.5**
- Mode = **9**

8. the following table would be plotted as (x,y) points, with the first column being the x values as number of mobile phones sold and the second column being the y values as money. To use the scatter plot for how many mobile phones sold.

x :4 1 5 7 10 2 50 25 90 36

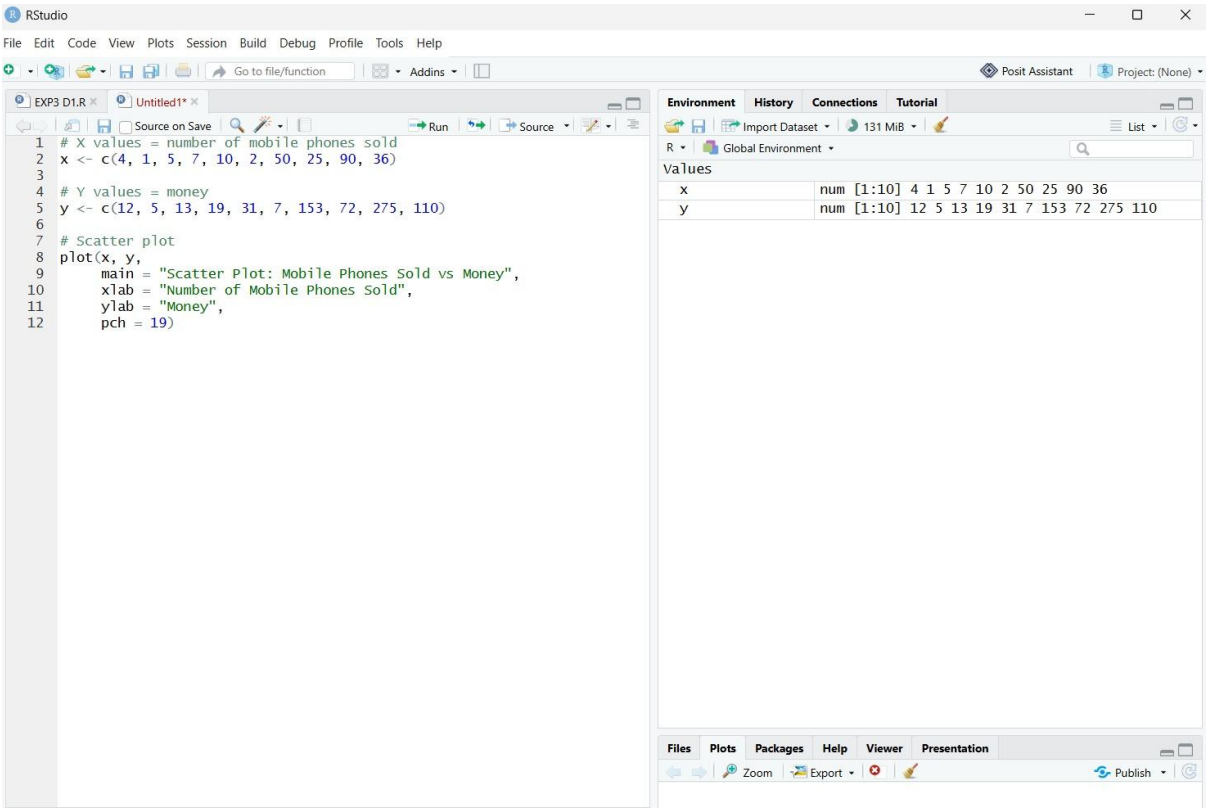
y :12 5 13 19 31 7 153 72 275 110

AIM:
To create a scatter plot showing the relationship between mobile phones sold and money.

ALGORITHM:

1. Create x and y vectors in R.
2. Assign mobile phones sold to the x-axis.
3. Assign money to the y-axis.
4. Use the plot() function.
5. Observe the relationship between the variables.

OUTPUT:



RESULT:
The scatter plot was successfully created and shows a **positive relationship** between mobile phones sold and money.

9. Implement of the R script using marks scored by a student in his model exam has been sorted as follows: 55, 60, 71, 63, 55, 65, 50, 55,58,59,61,63,65,67,71,72,75. Partition them into three bins by each of the following methods. Plot the data points using histogram.

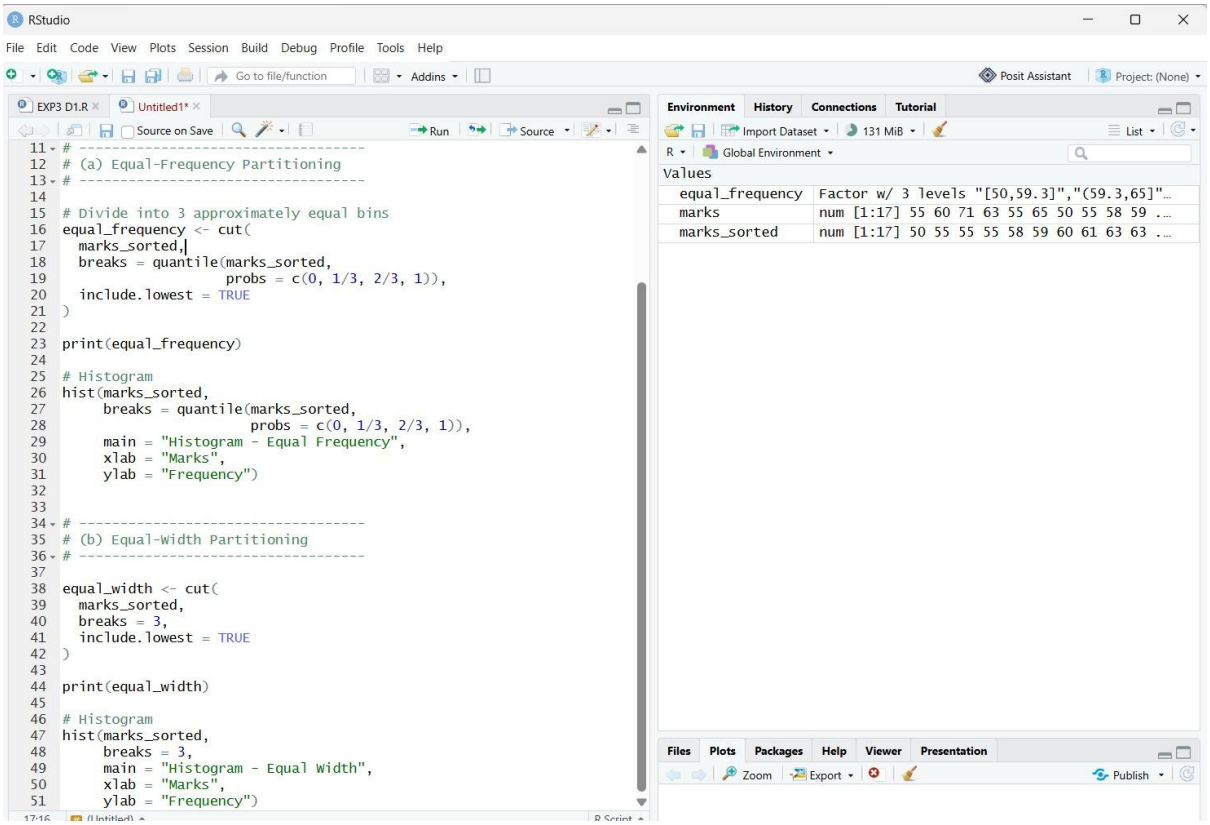
(a) equal-frequency (equi-depth) partitioning (b) equal-width partitioning

AIM:
To partition student marks into three bins using equal-frequency and equal-width methods and display them using a histogram.

ALGORITHM:

1. Enter the marks in R.
2. Sort the marks.
3. Divide the data into three equal-frequency bins.
4. Divide the range into three equal-width bins.
5. Create a histogram.
6. Display and compare the partitions.

OUTPUT:



RESULT:
The marks were successfully partitioned using **equal-frequency and equal-width methods**, and the histogram was successfully generated.

10. Suppose that the speed car is mentioned in different driving style.

Regular 78.3 81.8 82 74.2 83.4 84.5 82.9 77.5 80.9 70.6 Speed

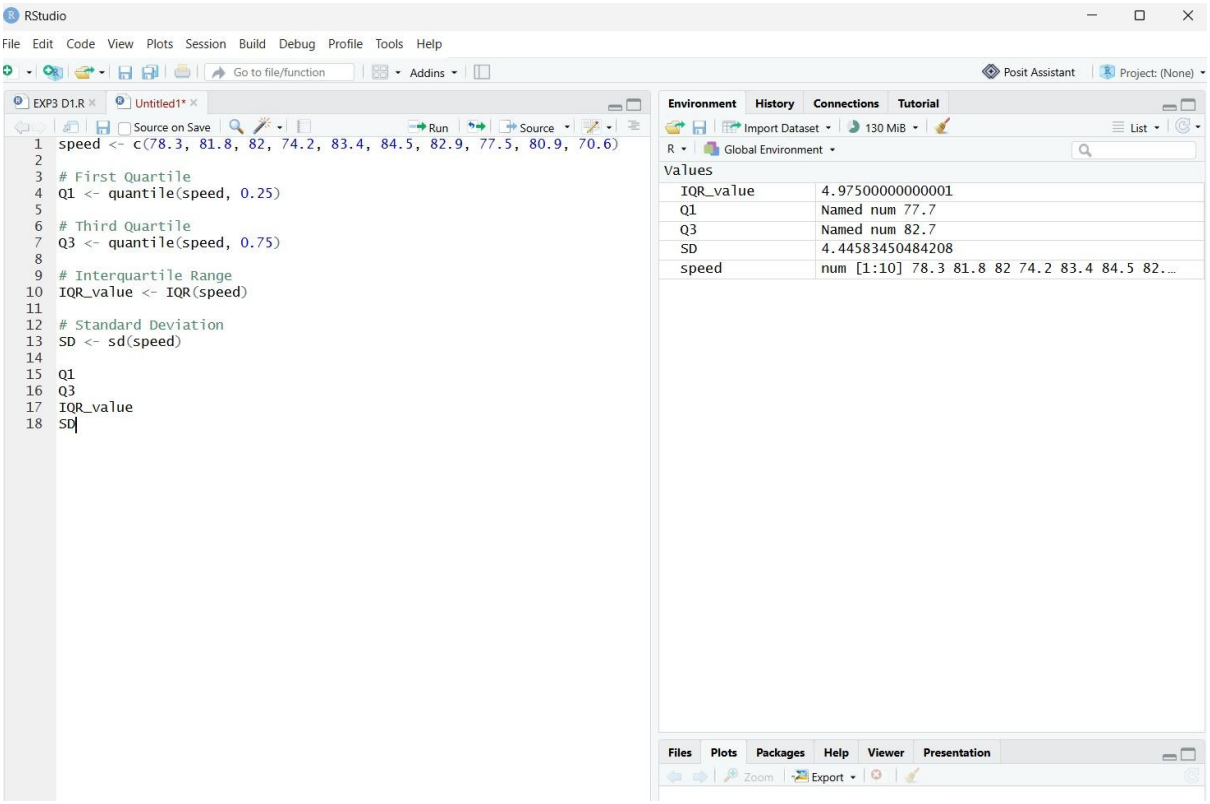
Calculate the Inter quantile and standard deviation of the given data.

AIM:
To calculate the interquartile range and standard deviation of car-speed data using R.

ALGORITHM:

1. Enter the speed values in R.
2. Calculate Q1 and Q3.
3. Calculate IQR using IQR().
4. Calculate standard deviation using sd().
5. Display the results.

OUTPUT:



RESULT:

- IQR = **4.975**
- Standard Deviation ≈ **4.446**

11. Suppose that the data for analysis includes the attribute age. The age values for the data tuples are (in increasing order) 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

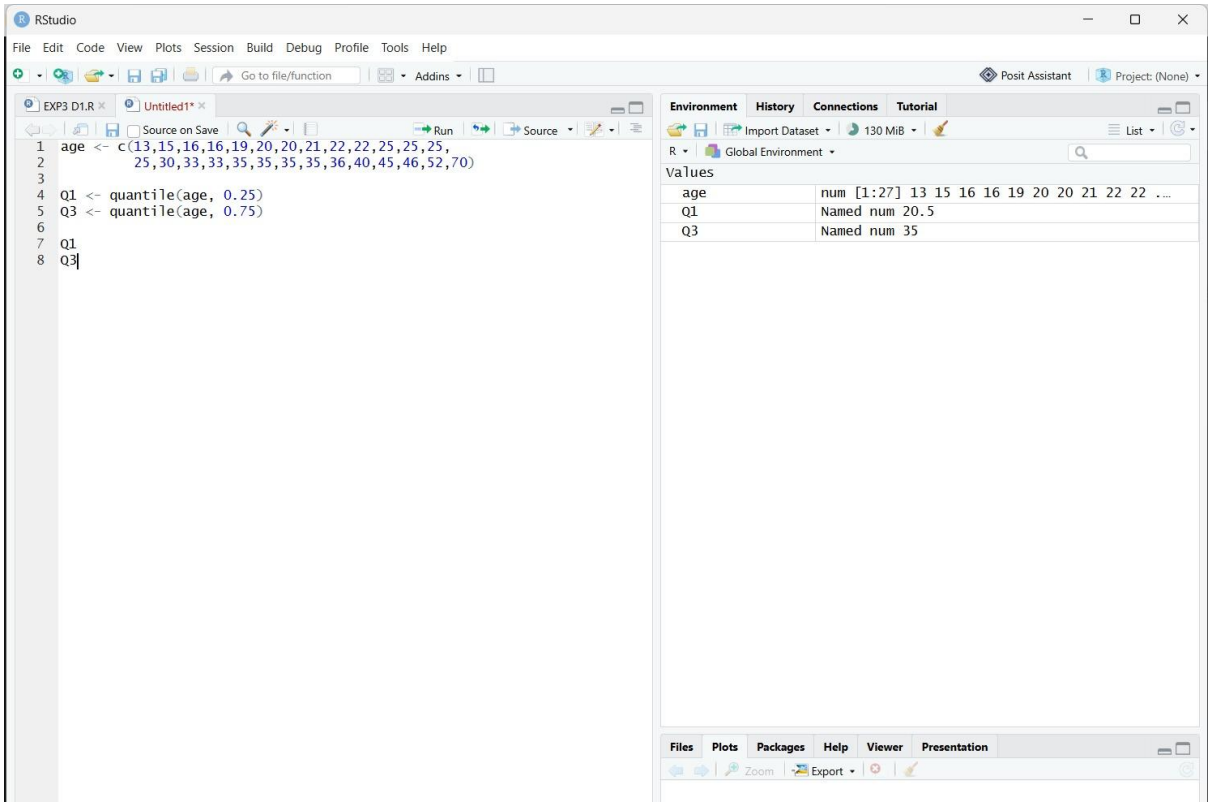
Can you find (roughly) the first quartile (Q1) and the third quartile (Q3) of the data?

AIM:
To calculate the first quartile (Q1) and third quartile (Q3) of the given age data.

ALGORITHM:

1. Enter the age values in R.
2. Arrange the values in ascending order.
3. Divide the data into lower and upper halves.
4. Find the median of the lower half as Q1.
5. Find the median of the upper half as Q3.
6. Display the quartile values.

OUPUT:



RESULT:

- **Q1 ≈ 20**
- **Q3 ≈ 35**